

Q1 We define $f(x, y)$ as number of different corresponding bits in the binary representation of x and y . For example $f(2, 7) = 2$, since the binary representation of 2 and 7 are 010 and 111. The 1st & 3rd bits differ so $f(2, 7) = 2$

You are given an array of n positive integers $A_1, \dots, A_n$. Find sum of $f(A_i, A_j)$ for all pairs (i, j) $1 \leq i < j \leq n$.
Return the answer modulo $10^9 + 7$

$$A = [1, 3, 5] \Rightarrow 8$$

$$A = [2, 3] \Rightarrow 2$$

Ans

$$1 = 001, 3 = 011, 5 = 101$$

$$= \begin{array}{r} 001 \\ 011 \\ \hline 101 \end{array} \text{ XOR } = \begin{array}{r} 010 \\ \hline \end{array}$$

$$\rightarrow \begin{array}{r} 010 \\ 101 \\ \hline 111 \end{array} \text{ XOR } = \begin{array}{r} 111 \\ \hline \end{array}$$

Now we ~~add~~ count the total

no. of 1s & multiply it

$$\text{by } 2 = 1 + 1 + 1 + 1 = 4 \times 2 = 8$$

Algorithm =

1) For all elements $1 - n$

2) calculate the XOR of $A[i]$ and $A[i+1]$ and store the result as r

3) ~~calculate~~ count the number of

- set bits in r and add it to ans.
- 3) Now in the next iteration set
 $r = \text{XOR of } A[i] \text{ and } r$
 - 4) Do the same for all elements
in A .

Algorithm $\Rightarrow$

- 1) ~~total sum~~ $\text{total} = 0$
- 2) $\text{bit} = 0 - 30$ (loop)
- 3) For each bit count elements in
array A where $(A[i] \& (1 \ll \text{bit})) = 1$
as c
- 4) Add $(2 * c * (n - c))$ to total
- 5) return total $\% (10^9 + 7)$